Submitting data to the Seafloor Data Repository (SDR)

NROC, MARCO, RWSC

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Background and Purpose

The Northeast Regional Ocean Council (NROC) and Mid-Atlantic Regional Council on the Ocean (MARCO) coordinate with providers and users of seafloor data to understand potential needs and applications.

NROC and MARCO convene working groups composed of decision makers and seafloor data experts in each region to continually advance the understanding of existing seafloor data, data gaps, data management workflows, and seafloor data products needed for ocean planning and management.

Federal agencies that collect seafloor data (or fund seafloor data collection) usually make those data available via federal data platforms, including:

- The International Hydrographic Organization Data Centre for Digital Bathymetry hosted at NOAA NCEI
- U.S. Geological Survey Coastal and Marine Hazards and Resources Program Imagery Data System | Samples Repository Collections Inventory
- Seafloor data collection funded by the National Science Foundation (NSF) is stored in the Marine Geoscience Data System

However, state agencies, other research entities, and partners in the private sector that collect seafloor data (or fund and/or require the collection of seafloor data) expressed a need for a platform to store, access, and transmit seafloor data for use in regional data products and future research. The recommendation for such a platform was also articulated in the RWSC Science Plan. NROC and MARCO developed a Seafloor Data Repository where data providers from multiple sectors could make data findable and accessible to meet these needs.

The purpose of this Seafloor Data Repository (SDR) is to catalog and make available seafloor datasets in U.S. Atlantic waters collected by a variety of entities and purposes that can eventually be used to develop regional-scale seafloor habitat data products (e.g., bathymetry maps, benthic habitat maps, surficial seafloor geology maps) for wider use and dissemination via the Mid-Atlantic Ocean Data Portal and Northeast Ocean Data Portal, and other mapping platforms.

Anyone may contribute data (or just metadata) to the SDR to provide for discoverability and accessibility of seafloor data in our regions of interest.

Please see the information in the Support tab for general guidance on publishing data.

Requiring submission to the Seafloor Data Repository

State and federal agencies who are funding research or otherwise requiring the collection of seafloor data are encouraged to require submission of seafloor data and data products to the Seafloor Data Repository. Use and/or modify the language below to include a requirement in a research grant/contract agreement or permit condition/stipulation.

> Contractor/permittee must submit data and data products (defined below) resulting from the activities described herein to the Seafloor Data Repository (https://seafloor.dataone.org) within 1 (one) year of collection.
>
> Contractor/permittee must follow the instructions and specifications in "Submitting Data to the Seafloor Data Repository (SDR)" available at: https://rwscollab.github.io/sdr
>
> Data should be submitted according to the specifications listed in Section 3 of "Submitting Data…" including by uploading the specific file types at the requested resolution. The resolutions requested are intended to obscure confidential business information, proprietary information, or sensitive seafloor features (e.g., cultural or national security). Contractor/permittee must provide agency/funder with a reasonable explanation for submission of data at different resolutions or any omission of data and data products due to concern about confidentiality or sensitivity.

Types of data accepted

The SDR stores data collected by any entity (federal, state, private company, research institution). Through funding from the Massachusetts Clean Energy Center, Rhode Island Department of Environmental Management, and U.S. Bureau of Ocean Energy Management, the following variables, formats, and resolutions were collaboratively identified by an NROC expert working group (INSPIRE and NROC 2023). These recommended variables, formats, and resolutions were selected and designed specifically to obscure potentially sensitive or confidential seafloor features (whether related to cultural resources or national security concerns).

Seafloor variable(s)	**Type/format**	**Desired resolution**
Multibeam bathymetry	Raster/tiff	8m*
Multibeam backscatter	Raster/tiff	8m*
Boulder fields and/or picks (identification of boulder locations)	Vector-points/polygons/ESRI shapefile	N/A
Seabed interpretation – CMECS Substrate Group, Subgroup	Vector-points/polygons/ESRI shapefile	N/A

Seafloor variable(s)	Type/format	Desired resolution
Seabed interpretation – Shell substrate	Vector-points/polygons/ESRI shapefile	N/A
Seabed interpretation – Bedforms	Vector-polygons/ESRI shapefile	N/A
Seabed interpretation – CMECS Biotic Subclass	Vector-polygons/ESRI shapefile	N/A

**Before uploading, bathymetry and backscatter mosaics should be down-sampled to the recommended resolution. A Down-sampling Protocol is available to assist data providers.*

Files containing confidential and/or sensitive information should not be uploaded to the Seafloor Data Repository. In these instances, metadata can be submitted to the Seafloor Data Repository, and the dataset entry can be labeled with the appropriate Data Sensitivity tag (see Steps for data submission).

Steps for data submission

1. Email emily.shumchenia@rwsc.org to let us know you're planning to submit a data package to the SDR.
2. You must be signed in to DataONE's platform to submit data. This can be done by entering your ORCID. If you don't have an ORCID, you can obtain one for free at https://orcid.org.
3. From the home page at https://seafloor.dataone.org, select "Submit Data" from the header menu.
4. The Submit Data page contains two sections. The top section is a tool to upload files. The bottom section is a data submission form.
5. Check that your metadata meets or exceeds the RWSC Essential Metadata guidelines. You will use this information to fill out the data submission form. The Seafloor Data Repository will automatically generate an EML (Ecological Metadata Language) metadata file using the information you submit. The metadata file will be available as part of your data record. *Note: EML is the metadata standard used by several RWSC/NROC/MARCO recommended repositories including OBIS and GBIF*.
6. Fill out all pages/subsections of the data submission form completely and accurately. Pay special attention to listing your research funder(s) with their identifiers (e.g. ROR) (use the *organization* and *userId* fields).
7. Based on the Data Sensitivity level that you assign to the data, upload data files using the tool at the top of the Submit Data page. If you select "Some or all data is sensitive with significant risk", **do not upload files** but continue to complete the metadata submission.

8. Hit the “Save dataset” button at the bottom of the page to send your data submission to the SDR curators for review and publication.

What to expect after submission

1. Your data remains private and will not appear in the SDR until SDR curators have reviewed it. SDR curators are:
    - Emily Shumchenia, RWSC Director, NROC Science & Portal Director (emily.shumchenia@gmail.com)
    - Nikelene Mclean, RWSC Coordinator, Habitat & Ecosystem Subcommittee Coordinator (mclean.nikelene@gmail.com)
2. During the review process, SDR curators may reach out to you with questions about the data or to discuss any sensitivities
3. Upon publication, curators will provide you with a DOI linked to the page specific to your dataset in the SDR. We encourage you to use the DOI to add the dataset to publication records, CVs, and websites, and to cite the data wherever they are mentioned to increase data discovery and reuse.
4. Once the data is published, it will appear in the SDR on the “Search Data” page. If you uploaded data, anyone can download it and the associated metadata. DataONE will track how many times a dataset has been downloaded, cited, and viewed. These metrics will be shown on your dataset’s SDR page.